



The Nexus Between Financial Inclusion and Renewable Energy Transition in Gulf Cooperation Council Economies: Implications for Sustainable Development

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Abstract: Financial Inclusion (FI) can have a significant role in Renewable Energy Transition (RET) in any region. This study investigates this nexus in the Environmental Kuznets Curve (EKC) framework of the resource-rich and geographically closed Gulf Cooperation Council (GCC) economies from 2000–2024. The Spatial Autoregressive (SAR) model is applied to this relationship due to economic, geographic, and policy interdependencies in the GCC region. RET is captured by Renewable Energy Output (REO) and consumption to analyze both demand- and supply-side proxies of the RET. The results show that RET in one economy has positive spillovers in the neighboring economies. However, FI negatively influences the RET in both proxies of the REO and Renewable Energy Consumption (REC). Thus, FI could not support the RET in the GCC region. Income per capita has a U-shaped effect on the RET. Therefore, economic growth can support the RET after a threshold point. Moreover, Foreign Direct Investment (FDI), Trade Openness (TO), and Human Capital (HC) promote the RET. The findings have significant implications for sustainable development in the GCC region. The results suggest that the GCC governments should support the financial sector in financing the renewable energy sector. Moreover, supporting globalization and HC development can enhance the RET. The positive spillovers also suggest regional coordination on sustainable energy policies.

Keywords: Financial Inclusion; Spatial analysis; Gulf Cooperation Council; Economic growth; Renewable Energy Transition

1. Introduction

The Renewable Energy Transition (RET) is a major policy priority for many economies due to rising global warming and the pressures of climate change. RET involves structural adjustment of sustainable energy systems, institutional reforms to support renewable energy, and changes in long-term investment patterns in favor of renewable energy (IPCC, 2022). Thus, RET would strengthen the sustainability framing with explicit reference to the Sustainable Development Goals (SDGs) and sustainability implications. For instance, the RET can support the target of affordable and clean energy (SDG 7). This SDG aims to shift from fossil fuel dependence to renewable and low-carbon systems (IPCC, 2022). In the case of the Gulf Cooperation Council (GCC) region, Saudi Vision 2030 and UAE Energy Strategy 2050 are designed to follow economic diversification for a sustainable energy future (Sanfilippo et al., 2024). However, the funding for the RET is still sourced from hydrocarbon rents, which seek to diminish this transition (Van de Graaf & Bradshaw, 2018).

Theoretically, inclusive financial systems can help in reducing transaction costs, mitigating risks, and mobilizing funds for green investments. Financial Inclusion (FI) can provide funds to households and Small and Medium Enterprises (SMEs) to adopt solar technology, which can help raise energy efficiency (Kor et al., 2025; Le et al., 2020). Thus, FI can channel savings and credit toward sustainable energy assets. However, FI may not inevitably redirect capital toward green energy and technologies in resource-dependent economies. Thus, FI may support carbon-intensive consumption and production in these economies. In this context, the empirical literature

has corroborated that the initial level of FI can raise environmental degradation in emerging economies (Parvez, 2021; Sadorsky, 2010). This dual expected effect of FI motivates us to inquire into this relationship in the resource-dependent GCC region. Within the GCC region, we also acknowledge the presence of heterogeneity that may affect the FI-RET relationship. Saudi Arabia and the UAE have initiated more ambitious renewable energy targets through Vision 2030 and Energy Strategy 2050, respectively. These initiatives may begin to redirect financial flows toward renewable projects. In contrast, Bahrain and Oman have smaller financial sectors and may face greater constraints in financing renewable transitions. This within-region heterogeneity motivates our inclusion of country-specific fixed effects. This also suggests that future research with longer post-reform data may reveal differential effects across member states.

The negative relationship between FI and RET is expected to be more dominant in the GCC region. In the rentier state structure, governments collect substantial revenues from hydrocarbon exports. This hydrocarbon revenue creates a path dependency and favors continued investment in established fossil-fuel infrastructure instead of renewable alternatives (Freer, 2020). Similarly, financial institutions are more experienced in financing hydrocarbon-based industries. Therefore, they are more likely to provide credit to the well-established hydrocarbon sector instead of the renewable energy sector. Moreover, most GCC countries provide fossil fuel subsidies. These subsidies can discourage renewable energy investments in the absence of comprehensive carbon pricing (Van de Graaf & Bradshaw, 2018). Therefore, FI may amplify existing production patterns rather than the RET. Moreover, GCC financial systems are highly centralized with state-owned banks, and financial access may follow government priorities. In the absence of clear policies to direct credit toward green investment, FI may support existing development patterns in favor of hydrocarbon investments.

The testing of FI on the RET is still missing in the GCC studies. In another methodological gap, the GCC economies are not independent of each other. These economies are highly spatially dependent due to geographical location, coordinated policies, and integrated financial markets. Thus, rising FI in one GCC economy could have spillover effects in another economy. Similarly, the literature has indicated the significance of conducting spatial analysis in integrated regions like the European Union, MENA, Sub-Saharan Africa (SSA), and ASEAN (Abbas et al., 2023; Elhorst, 2014; Sharif et al., 2024; Siddiqui & Iqbal, 2018). However, the application of spatial analysis in the FI-RET nexus is absent in the GCC literature. Figure 1 shows the spatial dimensions of the FI and the RET in the GCC region. FI indices, REC, and REO are spatially linked in the neighboring GCC countries.

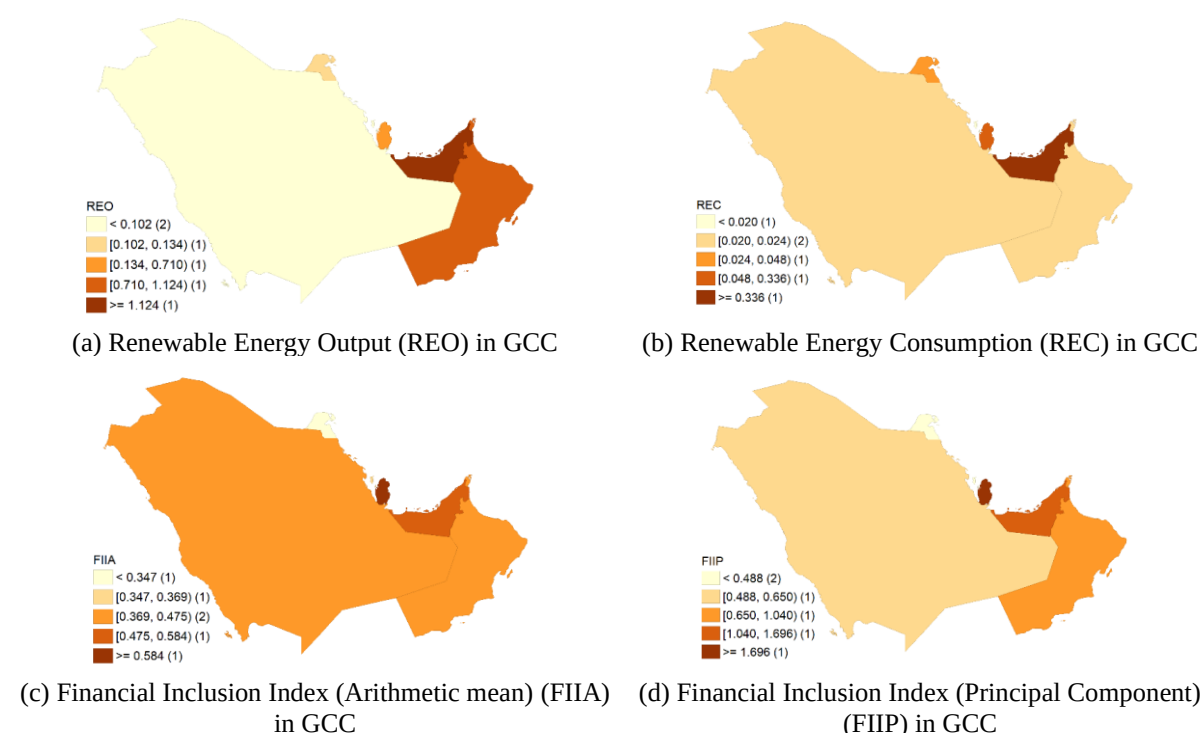


Figure 1. Distribution of the Renewable Energy Transition (RET) and Financial Inclusion (FI) indicators across Gulf Cooperation Council (GCC) countries (Period average: 2000–2024)

Considering the GCC literature gap and potential linkages in REC and REO, the study aims to investigate the influence of FI on REC and REO in the Environmental Kuznets Curve (EKC) framework of six GCC economies by using the data from 2000–2024 and by applying spatial econometrics. Moreover, Trade Openness (TO), Foreign Direct Investment (FDI), Human Capital (HC), and Corruption Control (CC) are also used as control variables in

the model to estimate the role of globalization and Institutional Quality (IQ) on the RET in the GCC region. The outcomes of this research would provide policy insights for FI in favor of the RET by considering the spatial linkages within the GCC region.

2. Literature Review

FI can reduce financial constraints in the way of the RET. For instance, FI can improve the capacity of resource-constrained SMEs to adopt renewable energy sources and energy-efficient technologies. However, FI can also accelerate the usage of fossil fuels in the absence of a financial regulatory structure. Therefore, the literature reports the mixed positive and negative effects of FI on the RET and environmental outcomes.

2.1 Financial Access Effects

The literature has corroborated the dual role of FI in determining the RET. A strand of literature has supported that FI can enhance green growth and environmental sustainability. Particularly, Digital Finance (DF) may enhance capital mobilization and reduce financial constraints. Therefore, DF can help in facilitating investments in renewable energy technologies. For instance, Chu et al. (2025) found that DF reduced financing barriers and improved resource allocation, which promoted REC in China. Similarly, Dong et al. (2025) investigated BRICS economies and found that FI improved green growth. Acheampong et al. (2025) extended the sample of the study by working on emerging economies and found that FI facilitated the RET in these economies. Sharif et al. (2024) found that FI mitigated emissions. However, population, globalization, and income amplified CO₂ emissions.

Shahbaz et al. (2022) investigated China from 2011–2017 and found that FI facilitated the reduction of emissions. This effect was achieved indirectly through lower energy consumption and an increase in the REC. Salakpi et al. (2025) investigated SSA and found that access to credit enabled households to buy solar panels and energy-efficient appliances. Moreover, financial depth helped firms to invest in large-scale renewable energy projects. Dai et al. (2022) found that sustainable FI served as an institutional and financial foundation to contribute to a greener economic structure. Shabir (2024) analyzed APEC countries from 2004–2018 and corroborated that FI and REC reduced emissions. Nevertheless, income and globalization amplified emissions. FI and emissions nexus was pleasantly moderated by technological innovation. Moreover, the collaboration of FI and globalization exacerbated emissions.

Conversely, some studies report that FI may initially increase carbon-intensive economic activities. Tian et al. (2025) explored emerging seven (E-7) nations and corroborated that FI supported energy-intensive industries. Thus, FI did not support the RET. Likewise, Zhong et al. (2024) reported that FI and financial deepening raised carbon-intensive activities and energy consumption. This effect was more pronounced due to the absence of environmental regulations and green finance frameworks. In the ecological effect of FI, Murshed et al. (2023) scrutinized emerging economies. FI raised energy usage and CO₂ in the household and business sectors by increasing demand for transportation and energy-intensive consumption. Anu et al. (2023) found that FI and industrial growth increased ecological degradation. This effect also increased in higher footprint quantiles. FI worsened environmental outcomes through structural and industrial channels.

Le et al. (2020) analyzed Asia from 2004–2014 and reported that FI, income, energy use, industrialization, urbanization, and FDI amplified emissions. Nevertheless, TO mitigated emissions. Makni & Delhoumi (2025) also reported that FI amplified emissions in the short run but decreased them in the long run in the 160 sample countries. However, this relationship was found to be inverted N-shaped in high-income nations and monotonic positive in low- to upper-middle-income nations. Moreover, REC and energy productivity reduced emissions in all income groups. Collectively, these studies highlighted that FI can increase environmental problems by increasing energy consumption.

This subsection shows the conditions under which FI shows positive or negative effects on RET. For instance, one group of studies argues that FI systems reduce transaction costs, reduce information asymmetries, and shift savings toward productive green investments (Chu et al., 2025; Dong et al., 2025). These mechanisms improve access to credit for solar panel adoption, energy-efficient appliances, and small-scale renewable projects (Salakpi et al., 2025). This renewable adoption helps in emission reduction through increased REC in China (Shahbaz et al., 2022), BRICS economies (Dong et al., 2025), and ASEAN countries (Sharif et al., 2024). However, in the absence of environmental regulations and green finance frameworks, FI accelerates emissions through energy-intensive consumption and fossil-fuel-based industrial production (Tian et al., 2025; Zhong et al., 2024) and increasing demand for transportation (Murshed et al., 2023). Thus, FI may facilitate or hinder the RET. The effect depends on the structural context in which financial access operates.

2.2 Resource Dependence Effects

The resource curse literature argues that heavy reliance on hydrocarbons reduces institutional development to

support the RET. Natural Resource Rent (NRR) was responsible for a lower level of FI (Badeeb et al., 2023; Bosah et al., 2024; Jie et al., 2024). NRR reduced governments' dependence on taxation and private-sector financing. This mechanism obstructed the financing for the RET. Similarly, Imran et al. (2025) found no nexus between NRR and economic progress. Governments' reliance on NRR reduced the focus on other economic activities and FI. Moreover, some studies corroborated that NRR and FI were responsible for higher ecological footprints and emissions in resource-abundant economies due to their energy-intensive extraction activities (Ali et al., 2022; Shang et al., 2025). Conversely, the literature also found that NRR served for sustainable development. For instance, Guan & Zhao (2024) found that the combined effect of FI and NRR reduced emissions by facilitating the reallocation of capital toward cleaner technologies. Similarly, Ma et al. (2023) corroborated that resource-rich economies mitigated environmental degradation when NRR was channeled by inclusive financial mechanisms. Thus, FI supported renewable energy investment and green infrastructure.

This subsection explains how natural resource abundance moderates the FI-RET relationship. The resource curse literature argues that heavy reliance on hydrocarbon rents reduces institutional development. The weak institutions reduce the diversification incentives for the RET in resource-abundant economies (Badeeb et al., 2023; Bosah et al., 2024). For instance, a government dependent on hydrocarbon revenues reduces the chance of introducing taxation systems and financial regulatory frameworks. These frameworks may channel financial resources toward RET. On the other hand, the literature also reports that a combination of effective governance and FI helps divert resource revenues towards green infrastructure investments (Guan & Zhao, 2024; Ma et al., 2023). The resultant green infrastructure may support RET and sustainable development in resource-rich economies. Thus, governance positively moderates the RET and FI nexus.

2.3 Institutional Moderation

The literature has also investigated the moderating role of IQ. For instance, Acheampong & Said (2024) found that rising financial access with stronger governance frameworks facilitated investment in cleaner technologies and energy sources, which improved energy efficiency. However, financial expansion raised carbon-intensive production and consumption in the presence of weak governance. Kassi et al. (2023) reported that governance quality improved the positive effects of FI on sustainable growth in the FI-renewable energy nexus. Thus, IQ helped align FI with sustainable energy objectives. At the regional level, Ali et al. (2022) found that effective economic governance moderated the effects of NRR and FI on ecological footprint. Thus, both NRR and FI reduced the ecological footprint in the presence of effective governance. These also helped reduce the resource curse effects. These studies collectively demonstrated that IQ frameworks significantly strengthened the role of FI in promoting sustainable growth. Zhou et al. (2024) investigated the MENA from 1990-2022. Environmental regulations positively moderated the nexus between financial development, FDI, and ecological footprint. Olaniyi et al. (2025) examined African economies from 1990-2019. These economies were found below a threshold of IQ to have positive environmental effects of financial development.

This subsection highlights that IQ positively moderates the relationship between FI and RET. For instance, a strong IQ facilitates investment in cleaner technologies (Acheampong & Said, 2024), reduces resource curse effects in the FI-RET nexus (Ali et al., 2022), provides strict environmental regulation to reduce ecological footprint (Zhou et al., 2024), and supports renewable energy sources (Kassi et al., 2023). These mechanisms help improve environmental sustainability. However, a weak IQ could not achieve the environmental benefits of financial development (Olaniyi et al., 2025).

2.4 Literature Gap and the Contribution of This Study

The reviewed literature signifies that FI can serve as a powerful lever for energy transition. However, this effect is conditional on IQ. Conversely, FI can increase environmental problems. This effect is found to be more pronounced in resource-abundant and fossil fuel-dependent economies. The GCC literature has not investigated this relationship. Moreover, spatial analysis in this relationship is mostly ignored in the global literature. Therefore, the study aims to probe the influence of FI on the RET in 6 GCC economies. This relationship is tested in the EKC framework by controlling for TO, FDI, HC, and CC in the model. Due to the geographically, environmentally, and economically interdependent economies of the GCC region, spatial analysis is also performed to estimate the spillovers of RET.

3. Methodology

3.1 Conceptual Framework

FI can facilitate the financing for the RET (Shahbaz et al., 2022). For instance, the financial channel suggests that FI can reduce transaction costs and mobilize savings toward green investments (Le et al., 2020). However,

this channel may direct capital toward fossil-fuel industries in the absence of green financial regulations (Ali et al., 2022; Shang et al., 2025). In the structural channel, hydrocarbon rents may reduce governmental dependence on taxation and private-sector financing in resource-abundant GCC economies. Thus, it reduces institutional incentives to diversify financial flows toward clean energy (Badeeb et al., 2023). In addition, the spatial channel recognizes that GCC economies are not isolated economies. The GCC economies are mostly coordinated by energy policies and integrated financial markets. Furthermore, the geographic proximity of GCC economies generates cross-border spillovers (Elhorst, 2014). These spillovers may also affect the nexus between FI and RET. Economic growth is a major driver of energy consumption, influenced by the level of economic activity. The EKC theory suggests a nonlinear association between income, energy consumption, and the ecological outcomes (Grossman & Krueger, 1991). Therefore, the nexus between FI and RET is tested in the EKC framework of the GCC economies.

3.2 Econometric Model, Data Sources, and Estimation Strategy

As per theoretical discussions, the model is hypothesized as follows:

$$ET_{it} = \beta_0 + \beta_1 FI_{it} + \sum_{k=1}^K \gamma_k Z_{k,it} + \mu_i + \lambda_t + \varepsilon_{it} \quad (1)$$

ET_{it} denotes the RET for country i in year t . It is captured by Renewable Energy Output (REO_{it}) and Renewable Energy Consumption (REC_{it}) to compare the results for both demand and supply aspects of the RET. In the independent variables, FI_{it} is measured using two indices to test the robustness. $FIIA_{it}$ is a simple average of institutional and market access. Moreover, $FIIP_{it}$ is also generated by applying Principal Component Analysis (PCA) on the same two dimensions. The PCA loadings, eigenvalues, explained variance, and construction diagnostics for the FIIP index are reported in Table A1 in the Appendix. In the PCA results, the first principal component had an eigenvalue of 1.52 and explained 76% of the total variance. Financial Institution Access (FIA) and Financial Market Access (FMA) were loaded positively on the retained component, with loadings of 0.82 and 0.57, respectively. Thus, a relatively greater contribution of FIA was given to the composite index. The correlation between FIIA and FIIP was found to be as high as 0.88. It suggests that both indices reflect the same broad concept of FI, although they differ in weighting structure. Thus, the alternative weighting structures preserve meaningful differences between the indices. This dual approach was followed to verify the robustness of the influence of FI on the RET. Both proxies were tested in separate models.

The vector $Z_{k,it}$ contains a set of control variables as per established models of sustainable development. The testing of the EKC is captured by the natural logarithm of Gross Domestic Product (GDP) per capita ($\ln YPC_{it}$) and its square ($\ln YPC_{it}^2$) as per theoretical predictions (Grossman & Krueger, 1991; Stern, 2004). TO (TOP_{it}) and FDI_{it} are taken as the globalization factors. These factors can help in technology diffusion and competitive pressures from globalization in adopting green energy sources (Pao & Tsai, 2011; Sadorsky, 2012; Siddiqui & Iqbal, 2018). Moreover, HC can develop the capacity to adopt renewable energy. Thus, HC_{it} is taken into the model. HC can help in developing, adopting, and managing advanced renewable technologies (Nelson & Phelps, 1966). Finally, control of corruption (CC_{it}) is taken as a proxy for IQ. It reflects policy effectiveness and investment security in any economy (Acheampong & Said, 2024; Kassi et al., 2023). λ_t and μ_i capture heterogeneity in time and cross-sections, respectively.

The hypothesized model in Eq. (1) may have spatial linkages in the geographically proximate GCC economies (Mahmood, 2022). Particularly, spatial linkages may be expected in the RET due to the diffusion of energy policies, technologies, and renewable practices across neighboring economies. Moreover, the GCC Secretariat actively coordinates energy policies and renewable targets. Thus, it may directly create policy transmission channels. For instance, Saudi Vision 2030 and UAE Energy Strategy 2050 are examples of the diffusion of energy policies. In addition, the interconnected electricity grids also show spatial linkages in energy infrastructure. Furthermore, the demonstration effect of successful renewable projects in one GCC economy can reduce perceived risks and lower adoption barriers in neighboring economies as well. These spatial linkages lead to spatial dependence in the dependent variable (RET), which can be modeled by Spatial Autoregressive (SAR) (Elhorst, 2014). However, spatial dependence may also operate through the Spatial Error Model (SEM). These linkages are expected due to omitted spatial variables such as common regional shocks and/or unobserved policy coordination (Anselin, 1988). In addition, spatial dependence may operate through independent variables, which can be estimated by the Spatial Durbin Model (SDM) (LeSage & Pace, 2009). For instance, FI in one GCC economy may affect the RET in neighboring economies through cross-border financial flows and policy diffusion. Therefore, the study estimated all spatial specifications and applied AIC and BIC criteria (LeSage & Pace, 2009). The analysis assessed the validity and suitability of the SAR in the hypothesized GCC model. Thus, the SAR model is applied to Eq. (2) in the following way:

$$ET_{it} = \rho \sum_{j=1}^N w_{ij} ET_{jt} + \beta_1 FI_{it} + \sum_{k=1}^K \gamma_k Z_{k,it} + \mu_i + \lambda_t + \epsilon_{it} \quad (2)$$

In Eq. (2), ρ is the SAR coefficient. w_{ij} is a matrix of the $N \times N$ dimension. It captures the spillovers of the RET in the GCC region. ϵ_{it} is a spatially uncorrelated error term. w_{ij} assumes a distance-based inverse matrix. Each element is defined as $1/d_{ij}$ for $i \neq j$. d_{ij} is the distance between the capital cities of GCC countries i and j . However, the diagonal elements of w_{ii} are zero. This matrix is also row-standardized to ensure the sum of the rows is 1. This procedure facilitates the interpretation of ρ as an average spillover effect from neighboring GCC economies.

The data for all variables mentioned in Table 1 were taken for the period 2000–2024. The data were sourced from the IMF (2025), the UNCTAD (2025), and the World Bank Group (2025). The definition and individual source of each variable are mentioned in Table 1.

Table 1. Description of variables and sources

Variable	Description and Construction	Source
REO _{it}	The total energy (kWh) generated from renewable sources.	World Bank Group (2025)
REC _{it}	The total consumption of energy derived from renewable sources, as a share of total final energy consumption.	World Bank Group (2025)
FIIA _{it}	Financial Inclusion Index (Arithmetic mean): A composite measure calculated as the arithmetic mean (ranging from 0 to 1) of two sub-indices from the IMF Financial Development Index: 1. FIA: Captures the depth and reach of traditional financial institutions (e.g., banks). 2. FMA: Captures the depth and reach of financial markets (e.g., stock markets). Formula: FIIA = (FIA + FMA)/2	IMF (2025)
FIIP _{it}	Financial Inclusion Index (Principal Component): An alternative composite index constructed on the same two sub-indices (FIA and FMA). This method assigns optimal weights to each sub-index based on their covariance. It maximizes the explained variance of the resulting composite indicator.	IMF (2025)
LnYPC _{it}	The natural logarithm of GDP divided by population. It is used to control for economic development and scale. GDP is in constant terms to remove inflation effects.	World Bank Group (2025)
TOP _{it}	A measure of an economy's integration into global trade, calculated as total trade divided by GDP.	World Bank Group (2025)
FDI _{it}	Net FDI Inflows (% of GDP).	World Bank Group (2025)
HC _{it}	Human Capital Index (UNCTAD): A composite index, which measures the stock of skills, knowledge, and health that a population possesses. This specific index is part of the UNCTAD's Productive Capacity Index, aggregating indicators related to education (e.g., school enrollment, literacy) and health (e.g., life expectancy).	UNCTAD (2025)
CC _{it}	It is a governance indicator that measures control of corruption and ranges approximately from -2.5 to +2.5.	World Bank Group (2025)

Note: REO = Renewable Energy Output; REC = Renewable Energy Consumption; FIIA = Financial Inclusion Index (Arithmetic mean); FIA = Financial Institutions Access; FMA = Financial Markets Access; FIIP = Financial Inclusion Index (Principal Component); LnYPC = Natural logarithm of GDP per capita; GDP = Gross Domestic Product; TOP = Trade Openness; FDI = Foreign Direct Investment; HC = Human Capital Index; CC = Control of Corruption; UNCTAD = United Nations Conference on Trade and Development. The data is publicly accessible from World Development Indicators (WDI):

<https://databank.worldbank.org/reports.aspx?source=2&series=IT.CEL.SETS.P2&country=WLD>; UNCTAD: <https://unctadstat.unctad.org/datacentre/dataviewer/US.PCI>; IMF: <https://data.imf.org/en/datasets/IMF.MCM.FDI>.

4. Empirical Results and Discussions

This section follows a sequence from data diagnostics to final estimations. Initial diagnostics are presented in Tables 2–4. These help to understand data properties and test the expected spatial linkages in the modelled variables. Then, the test for the spatial model selection is presented in Table 5. Later, the estimations of the baseline model with the dependent variable of REO are presented in Tables 6 and 7. Moreover, the robustness of these results is tested in Tables 8 and 9 with the proxy of REC as the dependent variable.

4.1 Preliminary Analysis and Diagnostic Tests

Table 2 shows the summary statistics of all variables. RET indicators (REO_{it} and REC_{it}) exhibit substantial positive skewness and high kurtosis. It shows cross-country and time variation in RET in the GCC panel. However, other variables show symmetric distributions with a moderate level of dispersion.

Table 3 shows pairwise correlation coefficients to test the issue of multicollinearity. A high correlation is found

between two proxies of the RET (REO_{it} and REC_{it}). Similarly, a high correlation is observed between two proxies of the FI ($FIIA_{it}$ and $FIIP_{it}$). However, high correlations do not exhibit the issue of multicollinearity. These variables are used in a separate econometric specification to test the robustness. Moreover, the low correlations observed among other variables confirm that the multicollinearity issue is not found in the hypothesized models.

Table 2. Summary statistics

Variable	Mean	SD	Skewness	Kurtosis	p25	p50	p75	Min	Max
REO_{it}	0.35	1.11	3.87	17.27	0.00	0.01	0.07	0.00	6.54
REC_{it}	0.07	0.21	4.49	23.26	0.00	0.00	0.10	0.00	1.30
$FIIA_{it}$	0.40	0.13	0.41	2.47	0.30	0.39	0.49	0.16	0.72
$FIIP_{it}$	0.64	0.75	0.39	2.33	0.00	0.59	1.13	-0.65	2.45
$LnYPC_{it}$	10.36	0.43	0.56	1.94	10.03	10.15	10.68	9.76	11.31
TOP_{it}	1.08	0.34	0.89	3.19	0.87	0.96	1.33	0.48	2.16
FDI_{it}	2.50	3.24	1.96	8.21	0.27	1.57	3.83	-2.76	17.25
HC_{it}	40.13	6.28	0.71	3.37	35.35	39.22	45.18	29.08	61.06
CC_{it}	0.45	0.44	0.21	2.13	0.14	0.37	0.80	-0.36	1.56

Note: REO = Renewable Energy Output; REC = Renewable Energy Consumption; FIIA = Financial Inclusion Index (Arithmetic mean); FIIP = Financial Inclusion Index (Principal Component); LnYPC = Natural logarithm of GDP per capita, GDP = Gross Domestic Product; TOP = Trade Openness; FDI = Foreign Direct Investment; HC = Human Capital Index; CC = Control of Corruption; SD = Standard Deviation; p25, p50 and p75 denote the 25th, 50th and 75th percentiles, respectively.

Table 3. Correlation matrix

	REO_{it}	REC_{it}	$FIIA_{it}$	$FIIP_{it}$	$LnYPC_{it}$	TOP_{it}	FDI_{it}	HC_{it}	CC_{it}
REO_{it}	1	-	-	-	-	-	-	-	-
REC_{it}	0.77	1	-	-	-	-	-	-	-
$FIIA_{it}$	0.07	0.13	1	-	-	-	-	-	-
$FIIP_{it}$	0.06	0.11	0.98	1	-	-	-	-	-
$LnYPC_{it}$	0.00	0.22	0.53	0.54	1	-	-	-	-
TOP_{it}	0.39	0.51	0.14	0.01	0.14	1	-	-	-
FDI_{it}	0.44	0.22	0.11	0.02	-0.12	0.43	1	-	-
HC_{it}	0.44	0.57	0.09	0.04	0.05	0.45	0.06	1	-
CC_{it}	0.20	0.39	0.52	0.52	0.67	0.31	0.10	0.02	1

Note: REO = Renewable Energy Output; REC = Renewable Energy Consumption; FIIA = Financial Inclusion Index (Arithmetic mean); FIIP = Financial Inclusion Index (Principal Component); LnYPC = Natural logarithm of GDP per capita, GDP = Gross Domestic Product; TOP = Trade Openness; FDI = Foreign Direct Investment; HC = Human Capital Index; CC = Control of Corruption. The symbol “-” omits redundant symmetric correlation entries.

In Table 4, spatial autocorrelation is tested on the non-spatial estimates. For this purpose, four models are estimated. Models 1 & 2 are estimated by assuming REO as a dependent variable with FIIA and FIIP as independent variables, respectively. The models also include other assumed independent variables. Similarly, keeping other things the same, Models 3 & 4 are estimated by assuming REC as a dependent variable. Moran’s I, LM-error, LM robust-error, LM-lag, and LM robust-lag strongly indicate the presence of spatial autocorrelations in the estimated non-spatial models. Thus, the estimated non-spatial models commit a bias of omitted spatial dimensions in the model. Therefore, the results of non-spatial models are biased. Consequently, the study proceeds to do spatial analyses for the hypothesized models.

Table 4. Detection of spatial autocorrelation

Tests	Model 1: FIIA & REO	Model 2: FIIP & REO	Model 3: FIIA & REC	Model 4: FIIP & REC
Moran’s I	20.654 (0.000)	6.573 (0.000)	12.541 (0.000)	9.215 (0.000)
LM-error	146.353 (0.000)	168.548 (0.000)	159.654 (0.000)	129.354 (0.000)
LM Robust-error	82.541 (0.000)	91.512 (0.000)	88.578 (0.000)	54.241 (0.000)
LM-lag	351.657 (0.000)	402.314 (0.000)	368.741 (0.000)	286.468 (0.000)
LM Robust-lag	144.519 (0.000)	186.496 (0.000)	177.249 (0.000)	133.333 (0.000)

Note: REO = Renewable Energy Output; REC = Renewable Energy Consumption; FIIA = Financial Inclusion Index (Arithmetic mean); FIIP = Financial Inclusion Index (Principal Component); LM = Lagrange Multiplier. *p*-value are in parentheses.

After confirming the spatial linkages, the SAR, SEM, SDM, and SAC specifications are applied to the baseline model. These specifications are tested by using the AIC and BIC criteria. The AIC and BIC values are found to be lowest for the SAR specification. Thus, SAR is considered the best spatial specification to run the hypothesized EKC model.

Table 5. Model comparison based on information criteria (AIC and BIC)

Model	AIC	BIC
SDM	312.442	339.538
SAR	284.383	328.553
SEM	316.869	343.965
SAC	313.823	339.929

Note: AIC = Akaike Information Criterion; BIC = Bayesian Information Criterion; SDM = Spatial Durbin Model; SAR = Spatial Autoregressive; SEM = Spatial Error Model; SAC = Spatial Autocorrelation Model.

4.2 Spatial Autoregressive Estimates for Baseline Model

REO is used as a dependent variable in Table 6 with FIIA as a proxy for FI and in Table 7 with FIIP as a proxy for FI. In both models, FI has a negative effect on REO. This negative coefficient suggests that increasing FI does not favor renewable energy production in the GCC region. This persistent negative relationship criticizes the theoretical positive relationship between FI and RET. For instance, access to credit may channelize capital toward sustainable technologies and energy sources. In comparison with the literature, the negative effect of FI on the RET is similar to the reported adverse environmental effects of FI in E-7 and emerging economies (Anu et al., 2023; Le et al., 2020; Murshed et al., 2023; Tian et al., 2025; Zhong et al., 2024). However, our finding contrasts with the evidence from China, ASEAN, BRICS, APEC, and SSA (Chu et al., 2025; Dong et al., 2025; Salakpi et al., 2025; Shabir, 2024; Shahbaz et al., 2022; Sharif et al., 2024), which validates that FI supports RET.

Table 6. SAR estimates of REO using FIIA

Variables	RE	CS FE	Time FE	CS and Time FE
FIIA _{it}	-1.841*** (0.685)	-1.298*** (0.366)	-1.911** (0.798)	-1.839** (0.809)
LnYPC _{it}	-25.451* (13.821)	-23.963* (13.944)	-39.655*** (13.289)	-65.778*** (14.433)
(LnYPC _{it}) ²	1.174* (0.657)	1.075* (0.597)	1.884*** (0.634)	3.143*** (0.688)
TOP _{it}	1.443*** (0.346)	1.437*** (0.355)	1.306*** (0.402)	1.200*** (0.298)
FDI _{it}	0.128*** (0.021)	0.131*** (0.020)	0.104*** (0.027)	0.149*** (0.020)
HC _{it}	0.063*** (0.012)	0.062*** (0.012)	0.088*** (0.031)	0.154*** (0.030)
CC _{it}	0.292 (0.210)	0.300 (0.207)	0.498 (0.339)	-0.282 (0.218)
ρ ($W \times \text{REO}_{it}$)	0.296*** (0.099)	0.285*** (0.099)	0.203** (0.101)	0.252** (0.121)
lgt_theta	-1.620*** (0.429)	-	-	-
σ^2_e	0.428*** (0.051)	0.407*** (0.047)	0.584*** (0.068)	0.308*** (0.035)
Constant	134.3* (72.49)	-	-	-
Observations	150	150	150	150
R-squared	0.365	0.296	0.438	0.356
Number of CS	6	6	6	6

Notes: SAR = Spatial Autoregressive Model; REO = Renewable Energy Output; W = Spatial weight matrix; σ^2_e = Error variance; FIIA = Financial Inclusion Index (Arithmetic mean). Standard errors are reported in parentheses. Model specifications include Random effects (RE), cross-sectional fixed effects (CS FE), time fixed effects (Time FE), and two-way fixed effects (CS and Time FE). ρ captures spatial dependence via the spatially lagged dependent variable $W \times \text{REO}$. lgt_theta denotes the logit-transformed spatial heterogeneity parameter, and σ^2_e is the error variance. The symbol “-” denotes omitted parameters for this model specification. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.01$.

The effect of economic growth shows a valid U-shaped Kuznets Curve with a negative sign of LnYPC_{it} and a positive sign of (LnYPC_{it})². Thus, REO is initially declining with rising economic growth. This is a scale effect of increasing hydrocarbon-based industries as per the typical industrial structure of the GCC economies. However, REO may rise after a threshold point at higher levels of development. This explanation is consistent with the income-effect hypothesis of Stern (2004). In Table 6, the estimated turning points are found at 50992, 69257, 37203, and 35038 constant US dollars in models 1, 2, 3, and 4, respectively. Qatar, Saudi Arabia, and the UAE have achieved this threshold in the sample period. However, other GCC countries' GDP per capita is still less than the threshold. FDI and TOP are taken as globalization factors in the estimations. Both factors are promoting the REO in the GCC region. This result follows the findings of Pao & Tsai (2011), which corroborated the positive

role of FDI in clean energy deployment in developing economies. Similarly, HC also improves REO. This finding is in line with the endogenous growth theory of Nelson & Phelps (1966). This theory highlighted the role of skills and knowledge in adopting and deploying renewable technologies. However, the effect of CC is insignificant. This result highlights that the weak CC measures could not support an energy-sector-specific regulatory framework to assist renewable energy projects. This result also contradicts the previous findings on the positive influence of IQ on RET and ecological quality (Ali et al., 2022; Kassi et al., 2023).

Table 7. SAR estimates of REO using FIIP

Variables	RE	CS FE	Time FE	CS and Time FE
FIIP _{it}	-0.540*** (0.133)	-0.622*** (0.128)	-0.273** (0.143)	-0.508*** (0.169)
LnYPC _{it}	-23.555* (13.801)	-22.067*** (9.812)	-38.191*** (13.263)	-62.853*** (14.565)
(LnYPC _{it}) ²	1.976*** (0.656)	0.980*** (0.255)	1.807*** (0.632)	2.000*** (0.495)
TOP _{it}	1.408*** (0.345)	1.388*** (0.352)	0.272*** (0.078)	1.195*** (0.399)
FDI _{it}	0.128*** (0.021)	0.131*** (0.020)	0.101*** (0.027)	0.149*** (0.020)
HC _{it}	0.063*** (0.012)	0.064*** (0.013)	0.104*** (0.032)	0.148*** (0.030)
CC _{it}	0.287 (0.208)	0.295 (0.204)	0.347 (0.247)	-0.243 (0.217)
ρ (W × REO _{it})	0.285*** (0.099)	0.273*** (0.099)	0.214** (0.107)	0.235** (0.110)
lgt_theta	-1.791*** (0.412)	-	-	-
σ^2_e	0.418*** (0.050)	0.398*** (0.047)	0.588*** (0.068)	0.308*** (0.036)
Constant	124.4* (72.422)	-	-	-
Observations	150	150	150	150
R-squared	0.326	0.263	0.396	0.361
Number of CS	6	6	6	6

Note: SAR = Spatial Autoregressive Model; REO = Renewable Energy Output; FIIP = Financial Inclusion Index (Principal Component); RE = Random Effects; CS = Cross-section; FE = Fixed Effects; W = Spatial weight matrix; ρ = spatial autoregressive coefficient; lgt_theta = logit-transformed spatial heterogeneity parameter; σ^2_e = error variance. Standard errors in parentheses; The symbol “-” denotes omitted parameters for this model specification. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A notable result in the models is a consistently positive coefficient of ρ . This result confirms that REO in one GCC economy has strong positive spillovers in other neighboring GCC economies through demonstration and diffusion effects. This result is in line with the spatial economic theory of cross-border effects (Elhorst, 2014; LeSage & Pace, 2009). Moreover, global energy prices and international climate policy developments are external pressures. GCC economies also share similar structural characteristics like resource dependence, rentier state structures, and comparable levels of economic development. These factors may lead to parallel policy responses to common challenges. In addition, oil price or investment shocks to one economy may transmit to others through trade and financial channels. These are not yet tested but may create similar responses in GCC economies.

These findings collectively underscore the critical importance of aligning financial and institutional frameworks with the broader objectives of the SDGs. The negative FI-RET nexus suggests that financial access alone is not sufficient to promote sustainability in the energy sector of resource-dependent GCC economies. On the other hand, the positive roles of FDI, trade, and HC suggest that sustainability transitions are multi-faceted. Thus, RET needs external technology transfers and internal capacity building for sustainable RET. The significant positive spatial spillovers further highlight that sustainable energy policies of one GCC economy may have cross-border effects. Thus, these spillovers can support regional energy sector sustainability.

4.3 Robustness Check with Renewable Energy Consumption as Dependent Variable

The robustness check is performed in this section by replacing the dependent variable of REO with REC in the models. The results are provided in Tables 8 and 9. Both proxies of FI, FIIA & FIIP, have negative effects on REC. However, the magnitudes of the effects are smaller than those of REO. The same observation is found in the case of all effects of other independent variables as well. Therefore, FI has the same effect on the demand-side proxy (REC) as of supply sided proxy (REO) of the RET. However, this finding contrasts with the existing literature, which documents a positive effect of FI on REC (Chu et al., 2025; Shahbaz et al., 2022). The U-shaped Kuznets

curve is validated in this model. It corroborates that economic growth increases REC after a threshold point. The globalization proxies, TOP & FDI, also have the same positive effects on REC. Thus, globalization is not only enhancing positive technological spillovers on REO but also enhancing awareness of cleaner energy consumption among the GCC economies. Similarly, HC is positively influencing the REC. Thus, education helps raise consumption sides of the renewable energy.

Table 8. Robustness check—SAR estimates of REC using FIIA

Variables	RE	CS FE	Time FE	CS and Time FE
FIIA _{it}	-0.769*** (0.130)	-0.836*** (0.125)	-0.428*** (0.125)	-0.301** (0.135)
LnYPC _{it}	-5.206** (2.382)	-5.523** (2.360)	-7.333*** (2.086)	-6.715*** (1.895)
(LnYPC _{it}) ²	0.251** (0.113)	0.289*** (0.102)	0.348*** (0.099)	0.315*** (0.090)
TOP _{it}	0.255*** (0.061)	0.246*** (0.060)	0.453*** (0.156)	0.311*** (0.052)
FDI _{it}	0.011*** (0.003)	0.012*** (0.003)	0.028*** (0.004)	0.0120*** (0.003)
HC _{it}	0.017*** (0.002)	0.018*** (0.002)	0.036*** (0.005)	0.048*** (0.004)
CC _{it}	0.107*** (0.035)	0.107*** (0.034)	0.147*** (0.038)	0.062*** (0.028)
ρ (W × REC _{it})	0.197** (0.106)	0.198** (0.105)	0.282* (0.156)	0.279** (0.125)
lgt_theta	-2.030*** (0.390)	-	-	-
σ^2_e	0.012*** (0.001)	0.011*** (0.001)	0.014*** (0.002)	0.005*** (0.001)
Constant	-16.512 (12.519)	-	-	-
Observations	150	150	150	150
R-squared	0.222	0.152	0.493	0.370
Number of CS	6	6	6	6

Note: SAR = Spatial Autoregressive Model; REC = Renewable Energy Consumption; FIIA = Financial Inclusion Index (Arithmetic mean); RE = Random Effects; CS = Cross-section; FE = Fixed Effects; W = Spatial weight matrix; ρ = spatial autoregressive coefficient; lgt_theta = logit-transformed spatial heterogeneity parameter; σ^2_e = error variance; R-squared = coefficient of determination. Standard errors in parentheses; The symbol “-” denotes omitted parameters for this model specification. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 9. Robustness check: SAR estimates of REC using FIIP

Variables	RE	CS FE	Time FE	CS and Time FE
FIIP _{it}	-0.142*** (0.022)	-0.153*** (0.021)	-0.038* (0.022)	-0.046** (0.022)
LnYPC _{it}	-4.629** (2.345)	-3.901* (2.318)	-7.341*** (2.058)	-6.274*** (1.899)
(LnYPC _{it}) ²	0.283** (0.121)	0.199* (0.110)	0.347*** (0.098)	0.293*** (0.091)
TOP _{it}	0.241*** (0.058)	0.233*** (0.059)	0.246 (0.611)	0.305*** (0.052)
FDI _{it}	0.011*** (0.003)	0.011** (0.003)	0.005*** (0.001)	0.012*** (0.003)
HC _{it}	0.017*** (0.002)	0.017*** (0.002)	0.039*** (0.005)	0.047*** (0.004)
CC _{it}	0.105*** (0.034)	0.105*** (0.034)	0.117*** (0.038)	0.164*** (0.028)
ρ (W × REC _{it})	0.288*** (0.105)	0.215** (0.103)	0.264* (0.156)	0.292** (0.135)
lgt_theta	-2.187*** (0.374)	-	-	-
σ^2_e	0.011*** (0.001)	0.011*** (0.001)	0.014*** (0.002)	0.005*** (0.001)
Constant	-18.784 (12.333)	-	-	-
Observations	150	150	150	150
R-squared	0.175	0.123	0.461	0.353
Number of CS	6	6	6	6

Note: SAR = Spatial Autoregressive Model; REC = Renewable Energy Consumption; FIIP = Financial Inclusion Index (Principal Component); RE = Random Effects; CS = Cross-section; FE = Fixed Effects; W = Spatial weight matrix; ρ = spatial autoregressive coefficient; lgt_theta = logit-transformed spatial heterogeneity parameter; σ^2_e = error variance; R-squared = coefficient of determination. Standard errors in parentheses; The symbol “-” denotes omitted parameters for this model specification. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Notably, the effect for CC becomes statistically significant on REC. This effect was insignificant in the case of REO. This result corroborates that institutions have developed better arrangements in enhancing the REC compared to the production side of renewable energy. For instance, GCC institutions have developed better standards for the consumption side of renewable energy. However, REO is a state-led initiative for renewable projects, which could not be influenced by institutional arrangements.

The robustness test validates the positive spillovers of REC among the GCC economies, like the positive spillovers of REO. Thus, the diffusion of the REC or REO is a regionally interconnected process within the GCC economies. Therefore, any increase in REC in one GCC country generates positive externalities for its neighboring countries as well. This result highlights the importance of the demonstration effect.

4.4 Robustness Check by Using an Alternative Weight Matrix

To test the robustness of all models, an alternative weight matrix of bilateral trade intensity has been applied to the baseline model of REC. This matrix captures economic integration through trade relationships. Bilateral trade also reflects the channels of policy diffusion and technology transfer from one GCC economy to another. Table 10 shows the results, and the signs and significance levels of all coefficients remain consistent with the estimations of the distance-based weight matrix. Moreover, the signs of the spatial coefficient (ρ) are also consistent in both weight matrices' specifications. Thus, RET exhibits positive spatial autocorrelation in the GCC region.

Table 10. SAR estimates using trade-based weight matrix

Variables	REO-FIIA Nexus	REO-FIIP Nexus	REC-FIIA Nexus	REC-FIIP Nexus
FIIP _{it}	-1.835** (0.807)	-0.511*** (0.171)	-0.305** (0.139)	-0.048* (0.026)
LnYPC _{it}	-66.112*** (14.125)	-63.214*** (14.684)	-6.748*** (1.935)	-6.301*** (1.961)
(LnYPC _{it}) ²	3.154*** (0.671)	3.008*** (0.351)	0.318*** (0.092)	0.296*** (0.102)
TOP _{it}	1.198* (0.682)	1.197*** (0.402)	0.313*** (0.056)	0.307*** (0.055)
FDI _{it}	0.148*** (0.019)	0.148*** (0.019)	0.012*** (0.001)	0.012*** (0.002)
HC _{it}	0.153*** (0.033)	0.149*** (0.029)	0.047*** (0.003)	0.046*** (0.004)
CC _{it}	-0.284 (0.220)	-0.245 (0.224)	0.063** (0.030)	0.165*** (0.030)
ρ (spatial)	0.248** (0.119)	0.231** (0.096)	0.276** (0.122)	0.289** (0.133)
Observations	150	150	150	150
R-squared	0.354	0.358	0.368	0.351
Number of CS	6	6	6	6

Note: SAR = Spatial Autoregressive Model; REO = Renewable Energy Output; REC = Renewable Energy Consumption; FIIA = Financial Inclusion Index (Arithmetic mean); FIIP = Financial Inclusion Index (Principal Component); LnYPC = Natural logarithm of GDP per capita; TOP = Trade Openness; FDI = Foreign Direct Investment; HC = Human Capital; CC = Control of Corruption; CS = Cross-section; ρ = spatial autoregressive coefficient. Standard errors in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Overall, the results provide insights that the FI reduces both REC and REO in the GCC region. However, HC helped improve both REC and REO. Moreover, FDI and TO are also external sources, which support both REC and REO.

5. Conclusion

FI has the potential to determine the RET in any economy. This research investigates the underexplored relationship between FI and RET in the EKC framework of the GCC region by using the SAR model. The results show that FI reduces the process of the RET by reducing both REO and REC. However, FDI and TO help increase both REO and REC. This result corroborates that globalization spreads positive spillovers by technology transfer and foreign pressure to adopt renewable energy sources. Moreover, HC positively impacts the REO and REC. Thus, increasing the stock of knowledge and skills in labor helps the RET in the GCC region. Furthermore, CC helps to raise REC but cannot affect REO. This result reflects that institutional frameworks more effectively support policies that promote REC. However, these could not perform well on the production side of renewable energy. Notably, the spillovers of the RET are found to be positive and significant. Thus, increasing RET in one GCC country motivates the neighboring GCC economies to raise the REC and REO. These spillovers are expected due to common renewable policies, knowledge diffusion, and shared infrastructure. Moreover, economic growth

shows a valid U-shaped effect on both REO and REC. Thus, economic growth after a threshold point would help in raising REO and REC in GCC economies.

The findings indicate a negative association between FI and the RET in GCC economies. To achieve the reverse situation, policy efforts should focus on improving the alignment between FI and RET objectives to promote sustainable energy transition. Given the positive role of FDI and TO, GCC countries should promote further globalization by strengthening their investment climate and trade linkages to attract foreign investments in renewable energy technologies. Moreover, given the role of HC in the RET, the government should further invest in education, technical training, and R&D with environmental awareness. Such government investment in skills, research, and knowledge development may support the RET by improving the region's capacity to adopt, operate, and expand renewable energy systems. The positive spatial spillovers of RET suggest that regional coordination on renewable energy development can increase the effects of national-level policies.

This study is constrained by data limitations in measuring RET. Therefore, REC and REO are employed as proxy indicators of RET. However, RET involves structural adjustment, institutional change, and long-term transformation of energy systems instead of just REC and REO. Future research may investigate more qualitative assessments of institutional change, policy stringency, and investment composition. This investigation may provide a more comprehensive understanding of the RET process in the GCC region. Moreover, the proposed models can be applied to the firm-level data to remove the aggregation bias. It will also be helpful in capturing heterogeneous effects from firm-level characteristics in the use of financial services. Moreover, the sample of the study can be expanded by working on the Middle East region. In addition, we only use CC as an IQ indicator. It is important for the general investment climate but does not capture the energy-sector-specific institutional arrangements. A future study may use energy-specific regulatory frameworks and governance measures to find the effects of these specific energy sector IQ indicators on the RET in the GCC region.

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Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The author declares no conflicts of interest.

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Appendix

Table A1. Principal component analysis for FIIP

Variables	PC1 Loading
FIA	0.82
FMA	0.57
Diagnostic	Value
Variables included	FIA, FMA
Number of variables	2
Components retained	1
Eigenvalue (PC1)	1.52
Variance explained (%)	76.0
Cumulative variance (%)	76.0
Loading signs	Positive for both variables
Correlation between FIIA and FIIP	0.88
Bartlett's χ^2	248.69

Note: FIIP = Financial Inclusion Index (Principal Component); FIIA = Financial Inclusion Index (Arithmetic mean); FIA = Financial Inclusion Access; FMA = Financial Market Access; PC = Principal Component.